SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I Shaik Abdul Adnan (192260084) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Q5

Total Marks 35 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ANSWERS – ANALYTICAL PROBLEM SOLVING

Question 1: Subnetting the 192.168.20.0/24 Network

Given network: 192.168.20.0/24. The network has 256 total addresses. The labs require /26, /27 and /28 subnets. The usable host count is calculated by 2^h − 2, where h is the number of host bits; two addresses are reserved for the network and broadcast addresses.

Lab A – /26: A /26 prefix leaves 32 − 26 = 6 host bits. The subnet mask is 255.255.255.192 and the block size is 256 − 192 = 64 addresses. Assign 192.168.20.0/26. Network address: 192.168.20.0. Valid host range: 192.168.20.1 to 192.168.20.62. Broadcast address: 192.168.20.63. Total addresses = 2^6 = 64; usable hosts = 64 − 2 = 62.

Lab B – /27: A /27 prefix leaves 5 host bits. The subnet mask is 255.255.255.224 and the block size is 32. The next correctly aligned subnet after Lab A is 192.168.20.64/27. Network address: 192.168.20.64. Valid host range: 192.168.20.65 to 192.168.20.94. Broadcast address: 192.168.20.95. Total addresses = 2^5 = 32; usable hosts = 32 − 2 = 30.

Lab C – /28: A /28 prefix leaves 4 host bits. The subnet mask is 255.255.255.240 and the block size is 16. Assign 192.168.20.96/28. Network address: 192.168.20.96. Valid host range: 192.168.20.97 to 192.168.20.110. Broadcast address: 192.168.20.111. Total addresses = 2^4 = 16; usable hosts = 16 − 2 = 14.

Therefore, the three non-overlapping subnets use 64 + 32 + 16 = 112 addresses in total. The remaining portion of the /24 block begins at 192.168.20.112. Each subnet boundary is aligned to its block size, so the labs can communicate within their assigned ranges without overlapping address allocations.

Question 2: Identifying the Subnets of Two Systems

Both systems use subnet mask 255.255.255.192. This mask is /26 because the first 26 bits are network bits. There are 6 host bits, so each subnet contains 2^6 = 64 total addresses and 64 − 2 = 62 usable host addresses. The block size in the last octet is 256 − 192 = 64.

The /26 subnet ranges in the 192.168.1.x network are .0–.63, .64–.127, .128–.191 and .192–.255. System 1, 192.168.1.130, falls in the .128–.191 block. Its network address is 192.168.1.128, valid host range is 192.168.1.129–192.168.1.190, and broadcast address is 192.168.1.191.

System 2, 192.168.1.190, also falls in the .128–.191 block. Its network address is 192.168.1.128, valid host range is 192.168.1.129–192.168.1.190, and broadcast address is 192.168.1.191. Both configured IP addresses are valid host addresses in this same subnet; .190 is the last usable host address, immediately before the broadcast address.

Conclusion: the two systems are on the same 192.168.1.128/26 subnet. Each subnet of this size supports 62 usable host addresses. Since both addresses belong to the same subnet, they can communicate directly at the IP subnet level, assuming their interface settings and other network configurations are correct.

Question 3: Department Subnet Allocation in 172.16.0.0/16

The allocated network is 172.16.0.0/16. The departments require different subnet sizes. For each prefix, the number of host bits is 32 minus the prefix length, and usable hosts equal 2^h − 2. To avoid overlap, each subnet is assigned at the next valid boundary after the previous allocation.

HR – /25: The mask is 255.255.255.128, leaving 7 host bits. Assign 172.16.0.0/25. Network address: 172.16.0.0. Valid hosts: 172.16.0.1–172.16.0.126. Broadcast: 172.16.0.127. Total addresses: 128; usable hosts: 126.

Sales – /26: The mask is 255.255.255.192, leaving 6 host bits. Assign the next aligned subnet, 172.16.0.128/26. Network address: 172.16.0.128. Valid hosts: 172.16.0.129–172.16.0.190. Broadcast: 172.16.0.191. Total addresses: 64; usable hosts: 62.

IT – /27: The mask is 255.255.255.224, leaving 5 host bits. Assign 172.16.0.192/27. Network address: 172.16.0.192. Valid hosts: 172.16.0.193–172.16.0.222. Broadcast: 172.16.0.223. Total addresses: 32; usable hosts: 30.

Admin – /28: The mask is 255.255.255.240, leaving 4 host bits. Assign 172.16.0.224/28. Network address: 172.16.0.224. Valid hosts: 172.16.0.225–172.16.0.238. Broadcast: 172.16.0.239. Total addresses: 16; usable hosts: 14.

The assigned subnets consume 128 + 64 + 32 + 16 = 240 addresses, from 172.16.0.0 through 172.16.0.239. The remaining unused range is 172.16.0.240 through 172.16.255.255 within the original /16 block. This is the continuous remainder after the four allocations; it can be divided into smaller subnets later as the organization grows.

Question 4: /27 Subnetting of 172.16.10.0/24

The original network is 172.16.10.0/24 and the new prefix is /27. Three bits are borrowed from the 8 host bits of the /24 network, so the number of subnets is 2^3 = 8. The /27 mask is 255.255.255.224, giving a block size of 256 − 224 = 32 addresses in the last octet.

Each /27 subnet has 32 total addresses. The number of host bits is 32 − 27 = 5, so usable hosts per subnet are 2^5 − 2 = 30. The eight subnet network addresses are 172.16.10.0, .32, .64, .96, .128, .160, .192 and .224.

To find the subnet for 172.16.10.78, compare the last octet with the 32-address blocks. The address 78 lies between 64 and 95, so it belongs to 172.16.10.64/27. Network address: 172.16.10.64. Valid host range: 172.16.10.65–172.16.10.94. Broadcast address: 172.16.10.95.

The address 172.16.10.95 is within the same 32-address subnet block, but it is the broadcast address, not a valid individual host address. Therefore, 172.16.10.78 is a valid host in this subnet, while 172.16.10.95 cannot be assigned to a device as its host IP address.

Question 5: IPv6 Address Compression and /64 Prefix

Given IPv6 address: 2001:0db8:0000:0000:0000:ff00:0042:8329. An IPv6 address contains eight groups of four hexadecimal digits, separated by colons. Compression removes leading zeros from each group and replaces one consecutive run of all-zero groups with a double colon (::). The double colon can be used only once in an address.

Compressed form: 2001:db8::ff00:42:8329. The leading zeros are removed from 0db8, 0042 and the other groups, and the consecutive all-zero groups are represented by ::. To expand it, restore the omitted zero groups until there are eight groups: 2001:0db8:0000:0000:0000:ff00:0042:8329. This matches the original address.

With prefix length /64, the first 64 bits (the first four hextets) form the network prefix. The network prefix is 2001:db8:0:0::/64, equivalent to 2001:0db8:0000:0000::/64. The remaining 64 bits are available for interface identifiers within this network.

The number of possible 64-bit interface identifiers is 2^64 = 18,446,744,073,709,551,616. Unlike common IPv4 subnet calculations, IPv6 does not use the same network-and-broadcast address subtraction rule; the full 2^64 interface-identifier space is the mathematical size of the /64 host portion. In practice, address assignment and reserved values depend on IPv6 configuration conventions.

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I Shaik Abdul Adnan (192260084) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Q5

25/30

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- Determine the network address and broadcast address for each department.
- Identify the valid host IP address range for each subnet.
- Calculate the number of usable host addresses available in each subnet.
- Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

- Determine the **network address** and **broadcast address** for each department.
- Identify the **valid range of host IP addresses**.
- Calculate the **number of usable hosts** in each subnet.
- Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- Convert the given IPv6 address into its **compressed notation**.
- Expand the compressed address back into its **full hexadecimal representation**.
- Identify the **network prefix** using the **/64** prefix length.
- Calculate the **total number of host addresses** supported within the subnet.
- the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

- The **destination subnet** to which the packet belongs.

- The **matching routing table entry**.
- The **next-hop network/interface** used to forward the packet.
- The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

- Determine the **network address** and **broadcast address** for each LAN.
- Identify the **valid host IP address range**.
- Suggest suitable IP addresses for **three computers** in each LAN.
- Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWERS – SCENARIO-BASED ASSESSMENT

Question 1: Smart Campus Subnet Allocation

Given network: 192.168.100.0/24. The subnets are allocated sequentially from the beginning of the address block.

1. School of Computing – 192.168.100.0/25

Network address: 192.168.100.0

Broadcast address: 192.168.100.127

Valid host range: 192.168.100.1 to 192.168.100.126

Usable hosts: 2^(32−25) − 2 = 128 − 2 = 126

2. School of Electronics – 192.168.100.128/26

Network address: 192.168.100.128

Broadcast address: 192.168.100.191

Valid host range: 192.168.100.129 to 192.168.100.190

Usable hosts: 2^(32−26) − 2 = 64 − 2 = 62

3. School of Mechanical Engineering – 192.168.100.192/27

Network address: 192.168.100.192

Broadcast address: 192.168.100.223

Valid host range: 192.168.100.193 to 192.168.100.222

Usable hosts: 2^(32−27) − 2 = 32 − 2 = 30

4. Remaining unused range: 192.168.100.224 to 192.168.100.255. This range contains 32 addresses (30 usable host addresses if treated as a /27 subnet).

Question 2: Company Department Subnet Allocation

Given network block: 10.10.0.0/16. The subnets are allocated sequentially from the beginning of the block.

1. Software Development – 10.10.0.0/24

Network address: 10.10.0.0; broadcast address: 10.10.0.255

Valid host range: 10.10.0.1 to 10.10.0.254

Usable hosts: 2^(32−24) − 2 = 254

2. Quality Assurance – 10.10.1.0/25

Network address: 10.10.1.0; broadcast address: 10.10.1.127

Valid host range: 10.10.1.1 to 10.10.1.126

Usable hosts: 2^(32−25) − 2 = 126

3. Human Resources – 10.10.1.128/26

Network address: 10.10.1.128; broadcast address: 10.10.1.191

Valid host range: 10.10.1.129 to 10.10.1.190

Usable hosts: 2^(32−26) − 2 = 62

4. Executive Management – 10.10.1.192/27

Network address: 10.10.1.192; broadcast address: 10.10.1.223

Valid host range: 10.10.1.193 to 10.10.1.222

Usable hosts: 2^(32−27) − 2 = 30

Unused range in the original /16 block: 10.10.1.224 to 10.10.255.255. The four assigned subnets use 480 addresses in total; the remaining 65,056 addresses are unallocated.

Question 3: IPv6 Addressing

Given IPv6 network: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed notation: 2001:db8:abcd::/64

2. Full hexadecimal representation: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. Network prefix: 2001:0db8:abcd:0000::/64. The first 64 bits (the first four hextets) identify the network prefix.

4–5. A /64 prefix leaves 64 bits for interface identifiers. Therefore, the subnet contains 2^64 = 18,446,744,073,709,551,616 possible interface addresses. IPv6 does not use the IPv4-style network and broadcast address subtraction; the exact number of addresses available for host/interface assignment can depend on network practices.

Question 4: Routing Table and Packet Forwarding

Destination IP address: 192.168.2.45

1. Destination subnet: 192.168.2.0/24, because the first 24 bits match that subnet.

2. Matching routing table entry: 192.168.2.0/24.

3. The router forwards the packet through the interface associated with the 192.168.2.0/24 route, or to that route's configured next hop. The question does not provide interface names or next-hop IP addresses, so a specific interface/IP cannot be determined from the given information.

4. If no specific routing-table entry matches the destination, the router uses the default route 0.0.0.0/0 if one is configured. If no default route exists, the packet is normally dropped.

Question 5: Administration and Finance LANs

1. Administration LAN – 192.168.10.0/26

Network address: 192.168.10.0

Broadcast address: 192.168.10.63

Valid host range: 192.168.10.1 to 192.168.10.62

2. Finance LAN – 192.168.10.64/26

Network address: 192.168.10.64

Broadcast address: 192.168.10.127

Valid host range: 192.168.10.65 to 192.168.10.126

Each /26 subnet has 2^(32−26) − 2 = 62 usable host addresses.

3. Example computer addresses:

Administration computers: 192.168.10.2, 192.168.10.3, and 192.168.10.4.

Finance computers: 192.168.10.66, 192.168.10.67, and 192.168.10.68.

4. Default gateways:

Administration LAN: 192.168.10.1

Finance LAN: 192.168.10.65

Each computer should use the gateway configured for its own LAN to communicate with devices on other networks.

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I Shaik Abdul Adnan (192260084) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Q5

25/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Name: Shaik Abdul Adnan | Reg No: 1972260084 | Course: CSA07 Computer Networks

Question 1: Workstation 192.168.10.65/26, gateway 192.168.10.1

Given: IP 192.168.10.65, mask 255.255.255.192 (/26), gateway 192.168.10.1. The host cannot reach systems in 192.168.10.0/26.

1. Subnet of the configured IP

/26 gives a block size of 256 − 192 = 64, so the subnets in 192.168.10.0/24 are .0, .64, .128 and .192. The address 192.168.10.65 lies in **192.168.10.64/26**.

2. Network, broadcast and host range

Subnet	Network	Broadcast	Valid host range	Usable
192.168.10.0/26	192.168.10.0	192.168.10.63	192.168.10.1 – 192.168.10.62	62
192.168.10.64/26 (workstation)	192.168.10.64	192.168.10.127	192.168.10.65 – 192.168.10.126	62

3. Validity of the IP and gateway

- **IP 192.168.10.65:** valid. It is the first usable host of 192.168.10.64/26 (not the network or broadcast address).
- **Gateway 192.168.10.1:** valid host, but it belongs to 192.168.10.0/26, a **different subnet** from the workstation. It is not directly reachable, so it is invalid for this host.

4. Configuration errors

- The workstation sits in 192.168.10.64/26, while the systems it must reach are in 192.168.10.0/26. Two different subnets need a router, and the host has no valid route.
- The default gateway is outside the host's own subnet. A host only ARPs for a gateway inside its own network, so the gateway is never reached and every packet to another subnet is lost.
- Root cause: the host IP and the gateway were taken from different /26 blocks.

5. Optimized configuration

Move the host into the target subnet 192.168.10.0/26 and keep the existing gateway:

Parameter	Corrected value
IP address	192.168.10.10
Subnet mask	255.255.255.192 (/26)
Default gateway	192.168.10.1

(Any free address from 192.168.10.2 to 192.168.10.62 works. 192.168.10.1 is reserved for the router.)

6. Usable hosts after correction

Host bits = 32 − 26 = 6, so usable hosts = 2^6 − 2 = 64 − 2 = **62** (192.168.10.1 to 192.168.10.62, of which one is used by the gateway).

Question 2: IPv6 devices A and B

Given: Device A 2001:db8::ff00:42:8329/64 and Device B 2001:db8:0:0:1::1/64. Both are on the same link but cannot communicate.

1. Full (expanded) notation

Device	Compressed	Expanded
A	2001:db8::ff00:42:8329	2001:0db8:0000:0000:0000:ff00:0042:8329
B	2001:db8:0:0:1::1	2001:0db8:0000:0000:0001:0000:0000:0001

In A, "::" replaces three zero groups. In B, "::" replaces two zero groups (groups 6 and 7).

2. Network prefix (first 64 bits = first 4 groups)

- Device A: **2001:0db8:0000:0000::/64** = 2001:db8::/64
- Device B: **2001:0db8:0000:0000::/64** = 2001:db8::/64

3. Same subnet?

Yes. With a /64 mask both addresses give the identical prefix 2001:db8::/64, so they are in the same subnet. The fault is therefore **not** a different prefix length or a different network prefix.

4. Addressing or prefix mismatch

- Device B's interface ID is 0001:0000:0000:0001. The "1" in the fifth group sits inside the interface-ID half, not in the subnet-ID field. This looks like a typing slip: the intended address was almost certainly **2001:db8:0:1::1** (subnet ID 1). Device B is placed in the wrong place in the addressing plan, and its interface ID is non-standard and hard to track.
- Device A's interface ID is 0000:ff00:0042:8329, which is also arbitrary and follows no planned pattern.
- Since the prefixes already match, the remaining causes of failure must be checked on the devices: IPv6 not enabled or no link-local address, wrong VLAN or port, a duplicate address, or ICMPv6/ND being filtered. Verify with the commands below.

```
Router(config-if)# ipv6 address 2001:db8:0:1::1/64   ! gateway
Router(config)# ipv6 unicast-routing
PC> ipv6 config            ! check address, prefix, link-local
PC> ping 2001:db8:0:1::20   ! test reachability
Router# show ipv6 interface brief
Router# show ipv6 neighbors  ! confirm Neighbor Discovery works
```

5. Optimized addressing

Device	Address	Prefix
Router (gateway)	2001:db8:0:1::1	/64
Device A	2001:db8:0:1::10	/64
Device B	2001:db8:0:1::20	/64

All three share the prefix 2001:db8:0:1::/64, with a clear, sequential interface ID in the last group.

6. Available host addresses

A /64 leaves 128 − 64 = 64 interface-ID bits, so there are 2^64 = **18,446,744,073,709,551,616** addresses. IPv6 has no broadcast address, so every address is usable (the all-zeros subnet-router anycast address is conventionally kept aside, giving 2^64 − 1 for hosts).

Question 3: Subnetting 192.168.1.0/24 with /26, /27, /28

Note: the question does not list the departments or their sizes, so the allocation and host requirements below are **assumed** to match the problem described (some departments short of addresses, others with many unused). The method applies to any real figures.

1. Possible subnets by mask

Mask	Block size	Subnets in /24	Usable hosts each	Network addresses (last octet)
/26	64	4	62	0, 64, 128, 192
/27	32	8	30	0, 32, 64, 96, 128, 160, 192, 224
/28	16	16	14	0, 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 224, 240

2. Assumed current allocation and requirements

Dept	Mask	Network	Broadcast	Host range	Usable	Needed	Unused / short
Engineering	/26	192.168.1.0	192.168.1.63	.1 – .62	62	50	12 unused
HR	/27	192.168.1.64	192.168.1.95	.65 – .94	30	10	20 unused
Sales	/28	192.168.1.80	192.168.1.95	.81 – .94	14	25	11 short
Admin	/28	192.168.1.96	192.168.1.111	.97 – .110	14	12	2 unused

3. Inefficient allocations

- **HR /27:** 30 usable for 10 hosts, so 20 addresses (about 67%) are wasted. A /28 is enough.
- **Sales /28:** only 14 usable for 25 hosts, so it is 11 addresses short. It needs a /27.
- **Engineering /26:** 12 spare addresses (about 19%). This is acceptable and allows growth.
- **Admin /28:** 2 spare, a good fit.

4. Allocation mistakes found

- **Overlap:** Sales 192.168.1.80/28 (.80 – .95) lies entirely inside HR 192.168.1.64/27 (.64 – .95). Overlapping subnets cause routing ambiguity and duplicate-address problems.
- **Wrong mask sizing:** subnets were chosen without matching masks to host counts (the big mask to the small department, the small mask to the large department).
- **No planning for WAN links:** router-to-router links would waste a full /28 or /27 each if added later.

5. Optimized VLSM scheme

Method: sort requirements from largest to smallest and allocate each at the next free block boundary, using the smallest mask that fits.

Dept	Hosts needed	Mask	Network	Broadcast	Host range	Usable	Spare
Engineering	50	/26	192.168.1.0	192.168.1.63	.1 – .62	62	12
Sales	25	/27	192.168.1.64	192.168.1.95	.65 – .94	30	5
HR	10	/28	192.168.1.96	192.168.1.111	.97 – .110	14	4
Admin	12	/28	192.168.1.112	192.168.1.127	.113 – .126	14	2
WAN link 1	2	/30	192.168.1.128	192.168.1.131	.129 – .130	2	0
WAN link 2	2	/30	192.168.1.132	192.168.1.135	.133 – .134	2	0

Free space for growth: 192.168.1.136 – 192.168.1.255 (120 addresses), one contiguous block, so no overlap and nothing wasted. Spare host addresses across the four departments change from 34 in the old plan (12 + 20 + 2, with Sales still 11 short) to 23 in the new plan (12 + 5 + 4 + 2), and every department now has enough addresses.

Check: Engineering .0–.63, Sales .64–.95, HR .96–.111, Admin .112–.127, WAN .128–.135. All blocks are aligned to their own size and do not overlap.

Question 4: IPv6 devices A and B

Given: Device A 2001:db8::ff00:42:8329/64 and Device B 2001:db8:0:0:1::1/64. Both are on the same link but cannot communicate.

1. Full (expanded) notation

Device	Compressed	Expanded
A	2001:db8::ff00:42:8329	2001:0db8:0000:0000:0000:ff00:0042:8329
B	2001:db8:0:0:1::1	2001:0db8:0000:0000:0001:0000:0000:0001

In A, "::" replaces three zero groups. In B, "::" replaces two zero groups (groups 6 and 7).

2. Network prefix (first 64 bits = first 4 groups)

- Device A: **2001:0db8:0000:0000::/64** = 2001:db8::/64
- Device B: **2001:0db8:0000:0000::/64** = 2001:db8::/64

3. Same subnet?

Yes. With a /64 mask both addresses give the identical prefix 2001:db8::/64, so they are in the same subnet. The fault is therefore **not** a different prefix length or a different network prefix.

4. Addressing or prefix mismatch

- Device B's interface ID is 0001:0000:0000:0001. The "1" in the fifth group sits inside the interface-ID half, not in the subnet-ID field. This looks like a typing slip: the intended address was almost certainly **2001:db8:0:1::1** (subnet ID 1). Device B is placed in the wrong place in the addressing plan, and its interface ID is non-standard and hard to track.
- Device A's interface ID is 0000:ff00:0042:8329, which is also arbitrary and follows no planned pattern.
- Since the prefixes already match, the remaining causes of failure must be checked on the devices: IPv6 not enabled or no link-local address, wrong VLAN or port, a duplicate address, or ICMPv6/ND being filtered. Verify with the commands below.

```
Router(config-if)# ipv6 address 2001:db8:0:1::1/64   ! gateway
Router(config)# ipv6 unicast-routing
PC> ipv6 config            ! check address, prefix, link-local
PC> ping 2001:db8:0:1::20   ! test reachability
Router# show ipv6 interface brief
Router# show ipv6 neighbors  ! confirm Neighbor Discovery works
```

5. Optimized addressing

Device	Address	Prefix
Router (gateway)	2001:db8:0:1::1	/64
Device A	2001:db8:0:1::10	/64
Device B	2001:db8:0:1::20	/64

All three share the prefix 2001:db8:0:1::/64, with a clear, sequential interface ID in the last group.

6. Available host addresses

A /64 leaves 128 − 64 = 64 interface-ID bits, so there are 2^64 = **18,446,744,073,709,551,616** addresses. IPv6 has no broadcast address, so every address is usable (the all-zeros subnet-router anycast address is conventionally kept aside, giving 2^64 − 1 for hosts).

Question 5: Routing table and longest-prefix match

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

1. Matching entry (longest-prefix match)

Destination 192.168.2.100 is compared against each route:

Route	Match?	Prefix length
192.168.1.0/24	No (third octet 2 ≠ 1)	–
192.168.2.0/24	Yes	/24
Default (0.0.0.0/0)	Yes (matches everything)	/0

The longest prefix is /24, so the entry **192.168.2.0/24** wins over the default route.

2. Next-hop router selected

10.0.0.3

3. Does the destination belong to the intended subnet?

Yes. 192.168.2.0/24 covers 192.168.2.0 – 192.168.2.255 (hosts .1 – .254), so 192.168.2.100 is a valid host in it. The forwarding decision is correct as written.

4. Configuration errors

The lookup itself is correct, so the problem lies in the route data rather than in the matching:

- The next hops of the two /24 routes are likely **swapped or wrong**: 10.0.0.3 may not be the router that actually connects to 192.168.2.0/24 (it may serve 192.168.1.0/24, or be down or unreachable).
- Next hops are given without an **exit interface** and without verifying that 10.0.0.2, .3 and .1 are on a directly connected network, so the next hop may not resolve through ARP.
- 10.0.0.3 may have **no return route** to the source, or its own routing may send the packet away from 192.168.2.0/24.
- If 10.0.0.3 is down, the router still prefers this route over the default route, so traffic is silently dropped and never falls back to 10.0.0.1.

5. Optimized routing table

Destination	Exit interface / next hop	Purpose
192.168.1.0/24	10.0.0.2 (verified)	Specific LAN 1 route
192.168.2.0/24	10.0.0.3 or the corrected router (verified)	Specific LAN 2 route
0.0.0.0/0	10.0.0.1	Single default route to the upstream
192.168.0.0/16 (optional)	Null0, high administrative distance	Discards unknown private traffic instead of sending it out to the Internet

- Verify each next hop with a ping and with "show ip route" and "traceroute", and fix the swapped or wrong next hop.
- Add an exit interface (or use a recursive next hop on a connected network) so the route is removed automatically when the link fails.
- Keep specific /24 routes; the two networks cannot be summarised into one route, since they use different next hops. If both later share a next hop, replace them with 192.168.0.0/22 or a suitable summary.
- For larger networks, use a dynamic protocol such as OSPF so failures reroute automatically.

```
Router(config)# no ip route 192.168.2.0 255.255.255.0 10.0.0.3
Router(config)# ip route 192.168.2.0 255.255.255.0 <correct next hop>
Router(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1
Router# show ip route
Router# ping 192.168.2.100
Router# traceroute 192.168.2.100
```

6. Route used if no destination matches

The **default route (0.0.0.0/0) via 10.0.0.1**, because /0 is the least specific match and is used only when no other prefix matches.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly

Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation